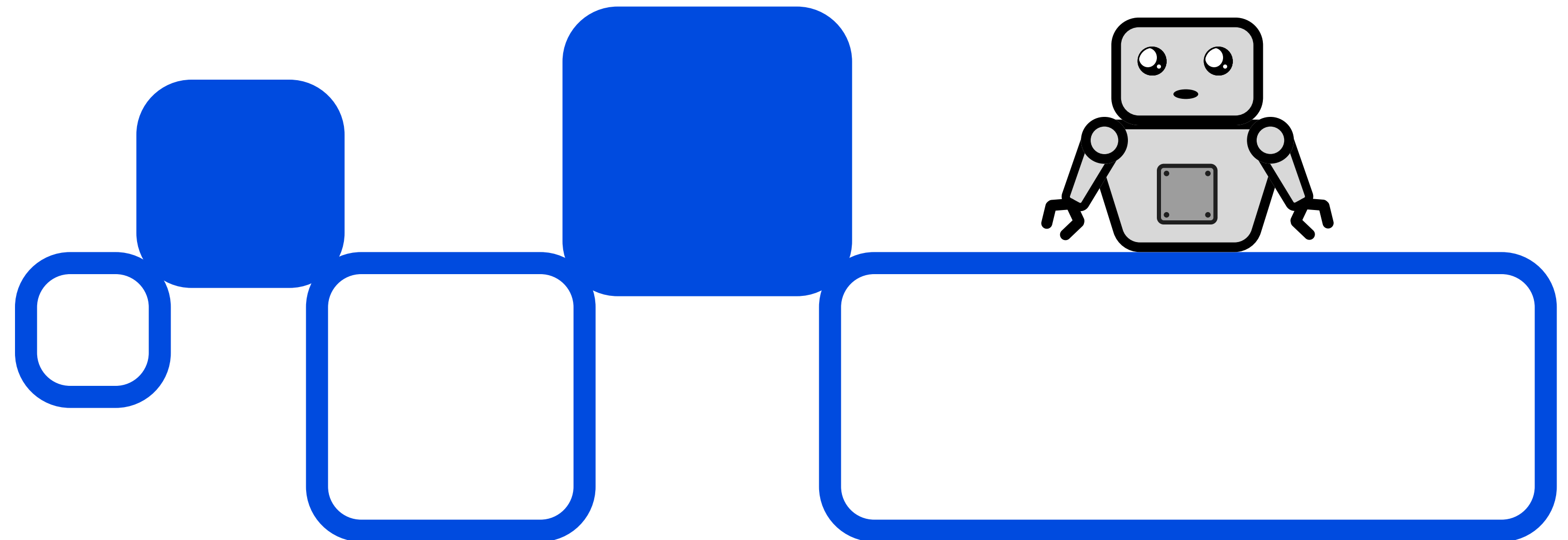
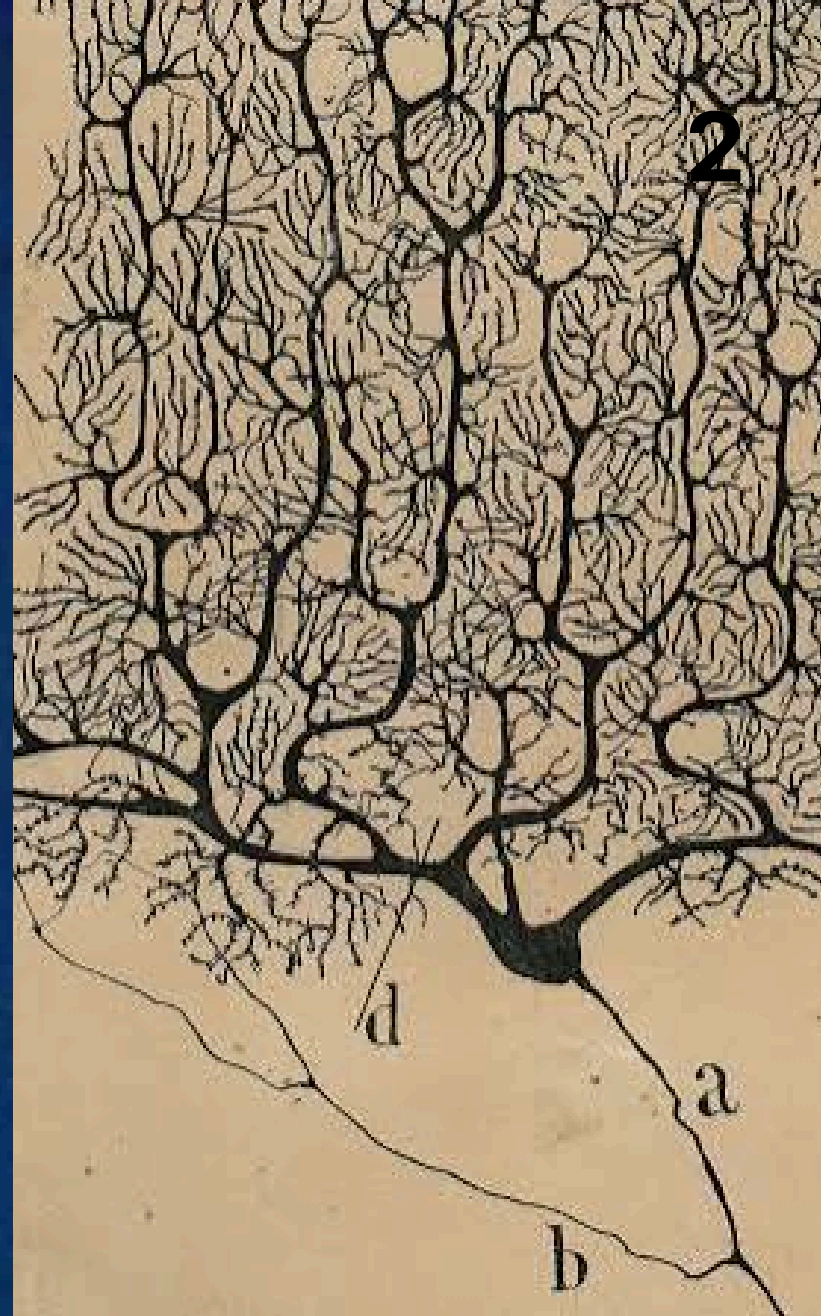




Universal Surprise-Minimizing Agents

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Presented by: Michele Vannucci

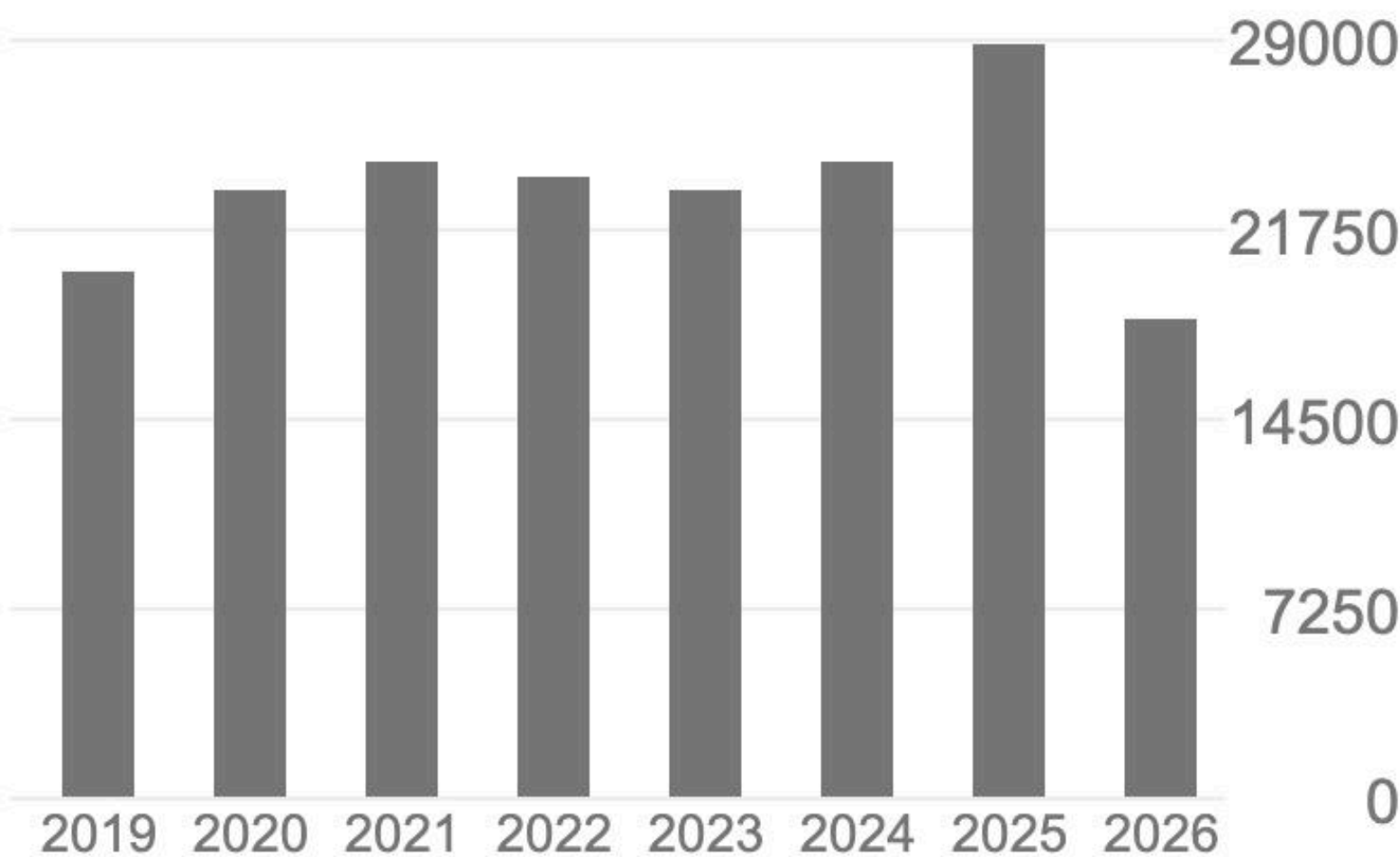




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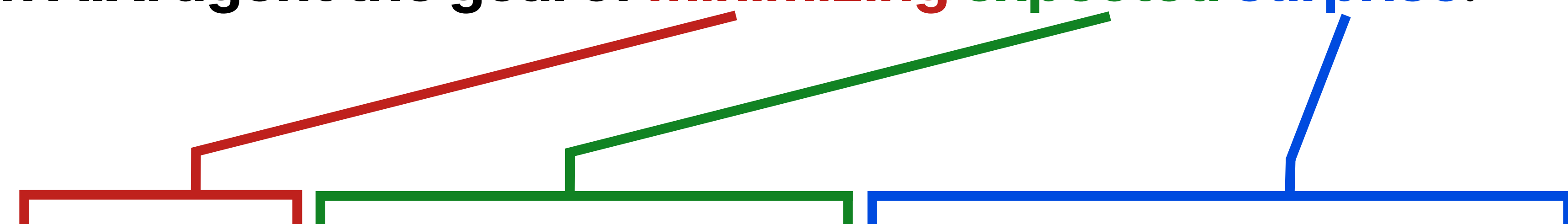
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Micro-abstract

We investigate whether the sole goal of surprise minimization on incoming observations leads to interesting or trivial behavior, within the computationally unconstrained setting of universal AI.

TL;DR:

We give an AI agent the goal of **minimizing expected surprise**.



$$a_{\text{SMA}} = \arg \min_{a_t} \sum_{o_t} \xi(o_t \mid h_{<t}, a_t) [-\log \xi(o_t \mid h_{<t}, a_t)]$$

Where: $h_{<t} := \langle a_1, o_1, a_2, o_2, \dots, a_{t-1}, o_{t-1} \rangle$

Why?

- The free energy principle describes surprise minimization as the overarching function of the brain
- There are only a few papers which study this objective within reinforcement learning formally
Here we consider the framework of universal AI to abstract away the computational boundedness confound variable
- On the other hand, within Universal AI, an opposite kind of intrinsic motivation, of entropy seeking, has been proposed with the Knowledge Seeking Agent (KSA)

Our hypothesis: exploratory behavior would emerge in complex environments

Background

- Intrinsic motivations provide a way to circumvent the issues related to externally defined rewards
- Some early work by Schmidhuber describes (human) curiosity as compression
- The Knowledge Seeking Agent (KSA) , inspired from scientific behavior, is shown to eventually learn the true environment (Orseau et al.)
- Nonetheless, asymptotically optimal agents are shown to eventually die or become incapacitated (Cohen et al.)
- The framework of universal AI facilitates discussion surrounding intrinsic motivations, since only a few parameters fully define the predictive model:

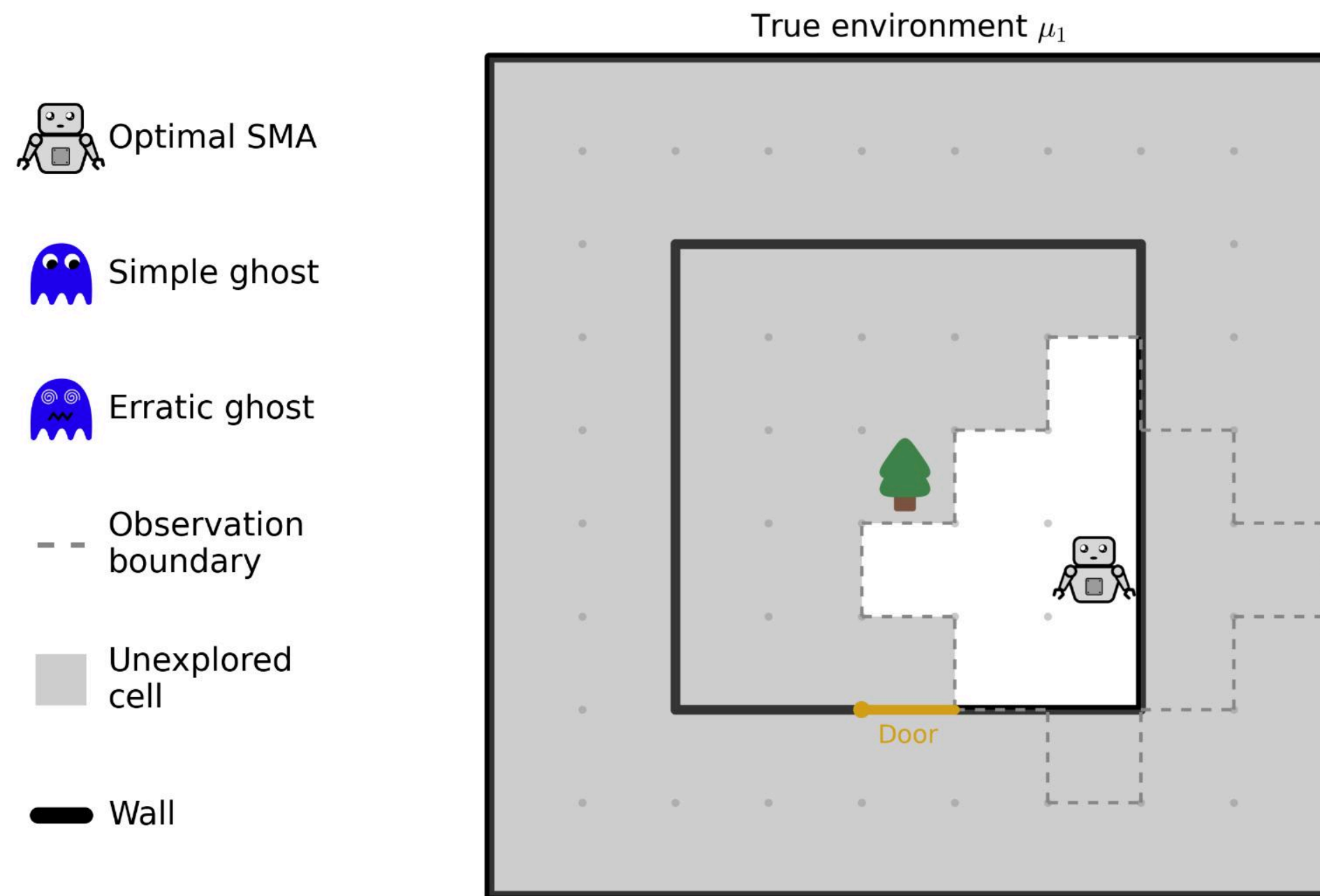
$$\xi(o_1 | a_1) = \sum_{\nu \in \mathcal{M}} w_\nu \nu(o_1 | a_1)$$

$\nearrow$ $\nearrow$ $\nearrow$
 $o_t \in \mathcal{O}$ $a_t \in \mathcal{A}$ $\nearrow$
 model class

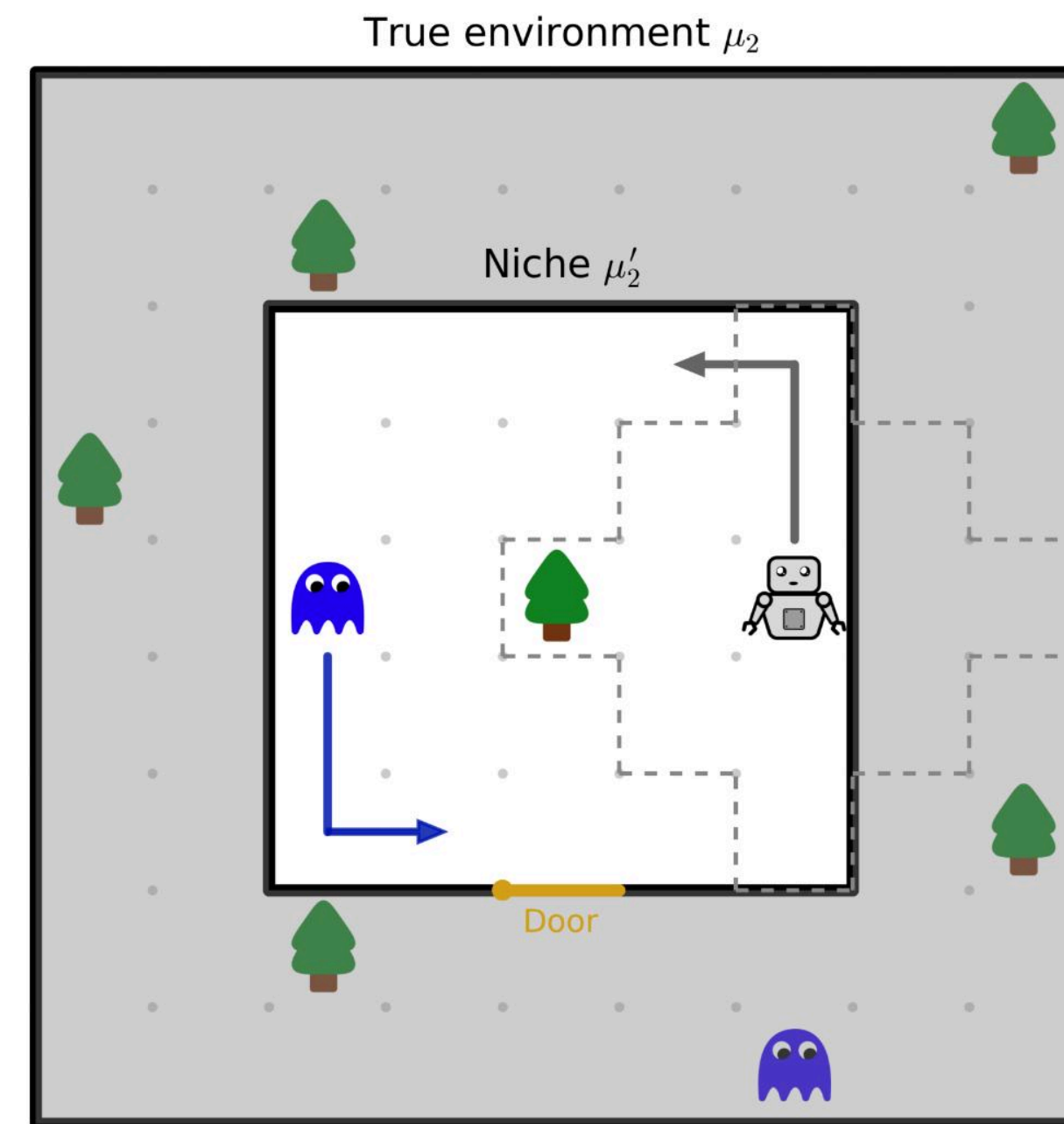
$$w_\nu := 2^{-K_U(\nu)}$$

$\uparrow$
 reference UTM

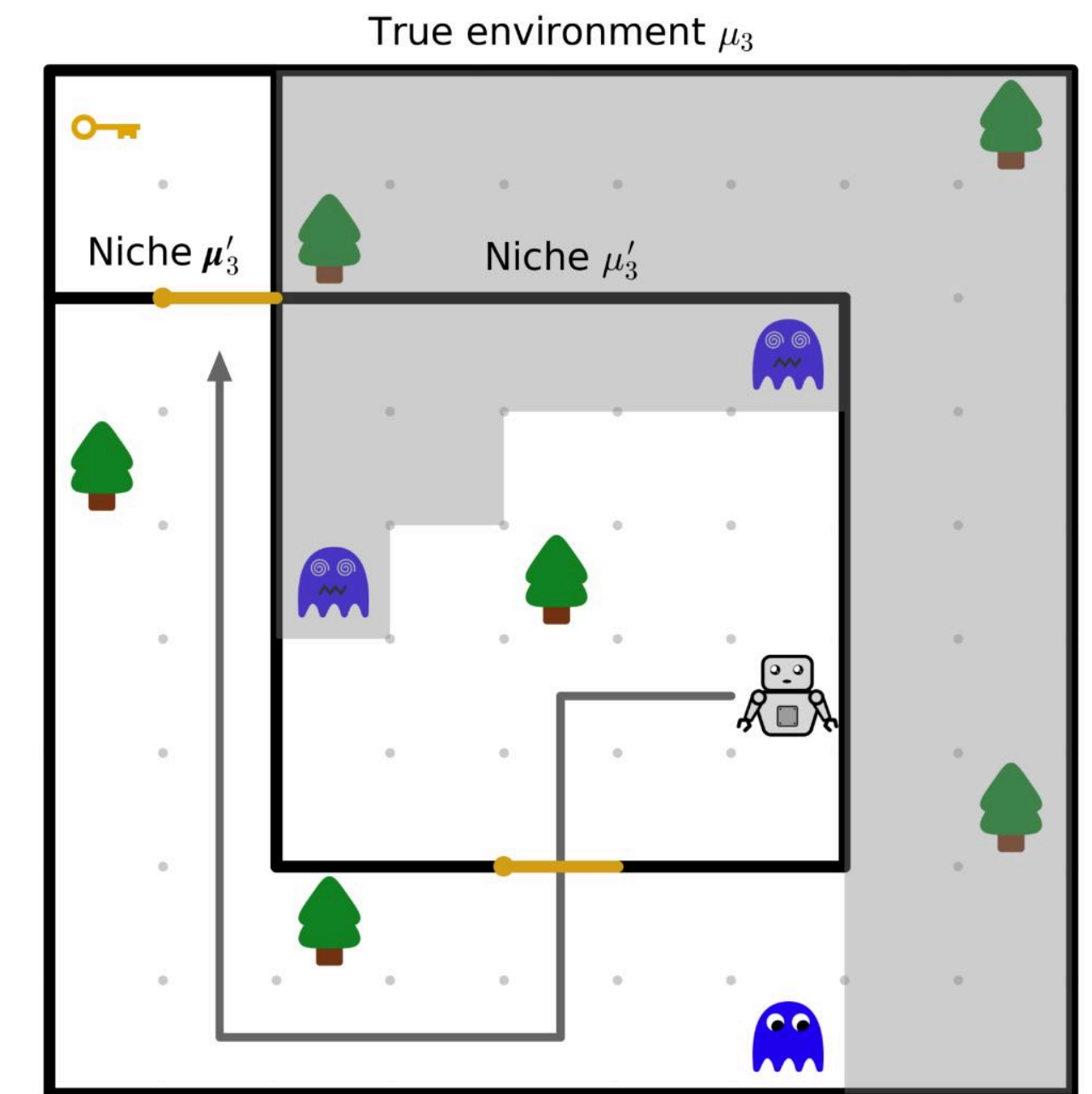
A Simple Example



(a) With no external stimuli, the agent does not move.



(b) While staying in the niche, the optimal behavior can be arbitrarily complex.



(c) The agent explores outside of the niche to minimize surprise in the long term.

Outline

~~Cost function derivation from FEP~~

Surprise and information gain

The oracle agent

- Minimal programs as niches
- Surprise minimization as *active* compression
- Escaping the dark room problem (and complex behavior)
- ~~Interactions between parameters~~

The agnostic agent

- Sublinear regret
- Optimality and death

Discussion

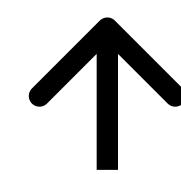
Definition (The Surprise-Minimizing Agent)

The agnostic agent

$$a_t \sim \pi_{\xi||\xi} = \arg \min_{\pi} S_{\xi||\xi}^{\pi, m}(h)$$

$$S_{\xi||\xi}^{\pi, m}(h) = \sum_{h' \in \mathcal{H}^{m-t}} P_{\xi}^{\pi}(h' | h) \left[-\log P_{\xi}^{\pi}(h' | h) \right]$$

$$P_{\xi}^{\pi}(h' | h) = \prod_{k=t}^m \xi(o_k | h_{<k} a_k) \pi(a_k | h_{<k})$$



The oracle agent

$$a_t \sim \pi_{\mu||\xi} = \arg \min_{\pi} S_{\mu||\xi}^{\pi, m}(h)$$

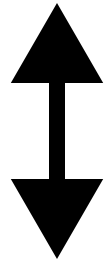
$$S_{\mu||\xi}^{\pi, m}(h) = \sum_{h' \in \mathcal{H}^{m-t}} P_{\mu}^{\pi}(h' | h) \left[-\log P_{\xi}^{\pi}(h' | h) \right]$$

Surprise and Information Gain

Key equality: Surprise = information gain + aleatoric uncertainty

$$S_{\xi||\xi}^{\pi}(h) = \underbrace{\mathbb{E}_{\xi}^{\pi}[IG \mid h]}_{\text{information gain}} + \underbrace{\sum_{\nu \in \mathcal{M}} w_{\nu}(h) S_{\nu||\nu}^{\pi}(h)}_{\text{ambiguity}}$$

Equivalent in the
deterministic setting



	Maximization	Minimization
IG	KL-KSA	KMA
Surprise	Shannon-KSA	SMA

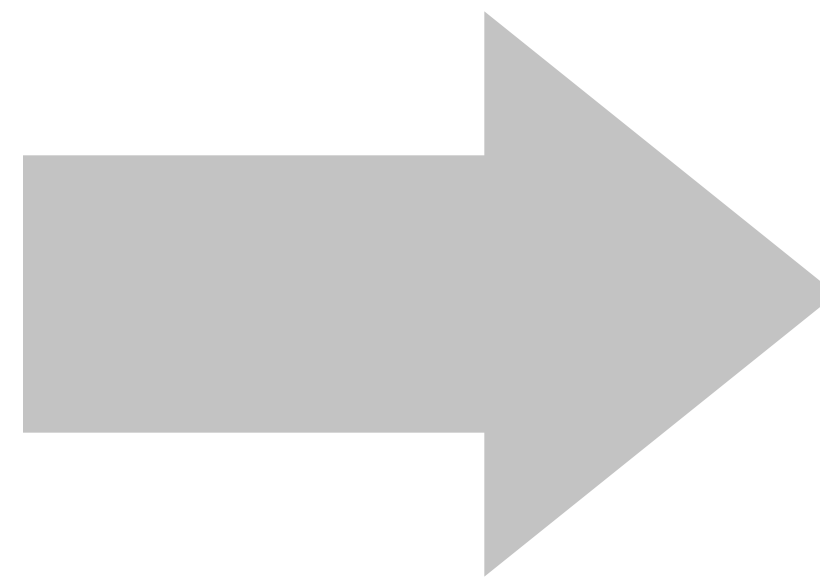
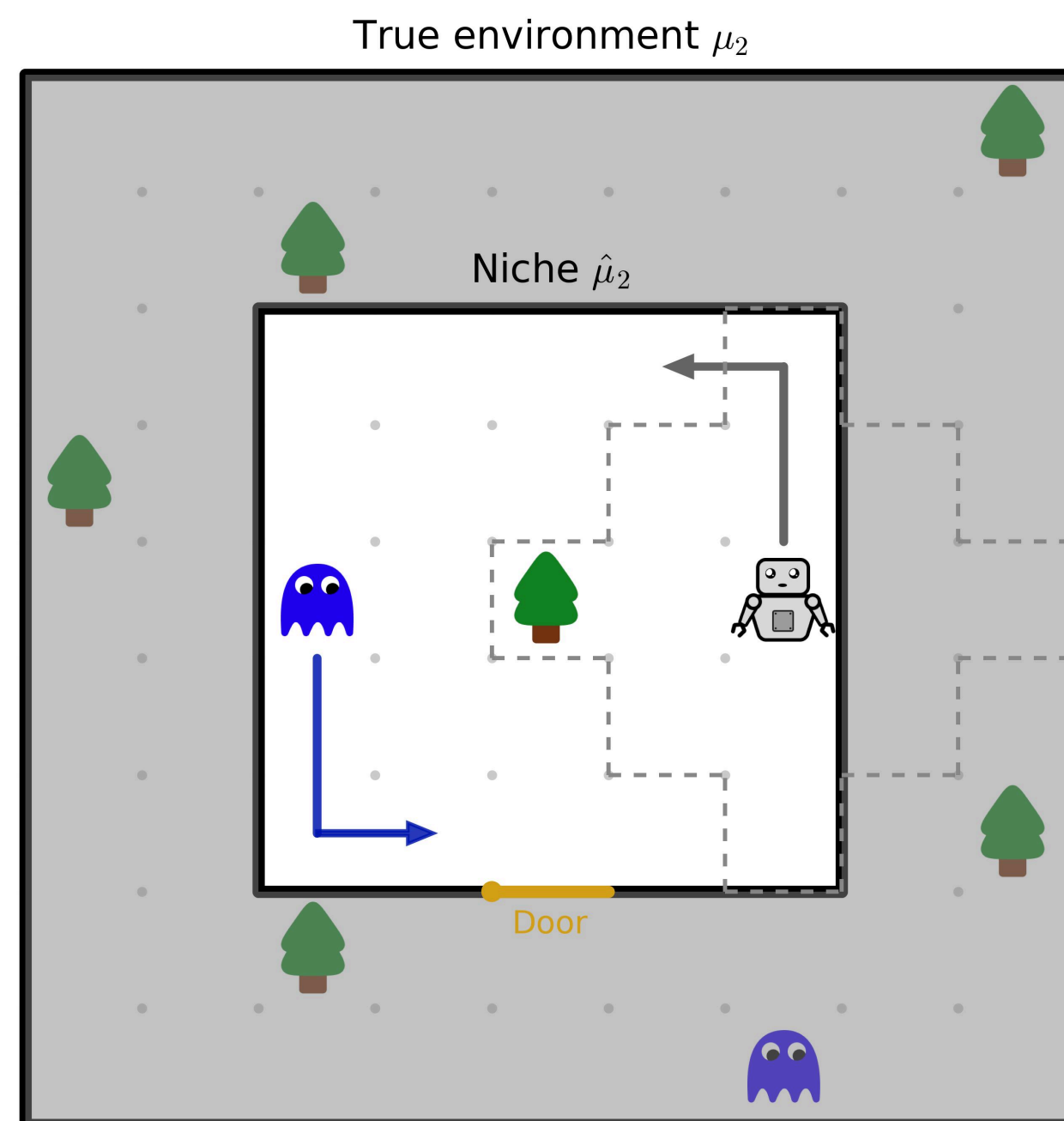


Niches

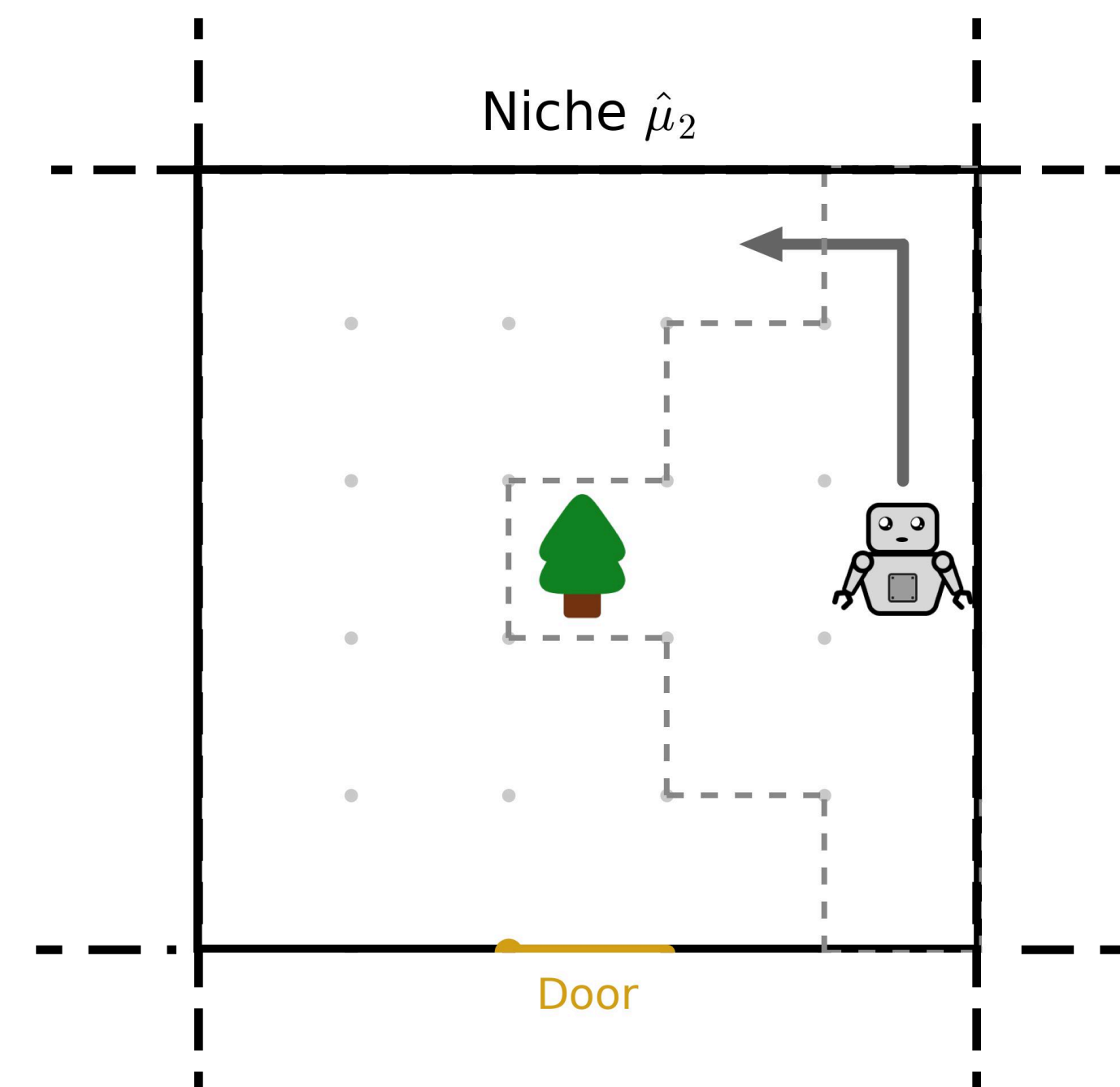
An environment $\hat{\mu}$, belonging to the class of deterministic environments $\mathcal{M}$, is a π -niche of an environment $\mu \in \mathcal{M}$ if they produce the same observations under policy π :

$$\hat{\mu} \in \mathcal{M}_{\mu}^{\pi, m}(h) := \left\{ \nu \in \mathcal{M} \mid \exists h' \in \mathcal{H}^m : P_{\nu}^{\pi}(hh') = P_{\mu}^{\pi}(hh') \right\}$$

The true environment



What the agent “believes”

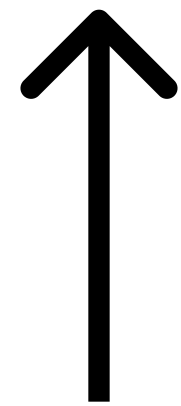


Fundamental Relationships

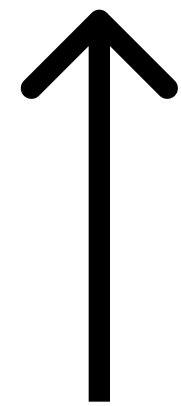
with $\mathbf{o}, \mathbf{a}$ in $\mathbf{h}$ and $\xi(\mathbf{o} \parallel \mathbf{a}) = \prod_{k=0}^m \xi(o \mid h_{<k} a_k)$

$$S_{\mu \parallel \xi}^{\pi^*, m}(\epsilon) = \sum_{\mathbf{h} \in \mathcal{H}^m} P_{\mu}^{\pi^*}(\mathbf{h}) [-\log \xi(\mathbf{o} \parallel \mathbf{a})] = -\log \left[\sum_{\nu \in \mathcal{M}_{\mu}^{\pi^*, m}} w_{\nu} \right]$$

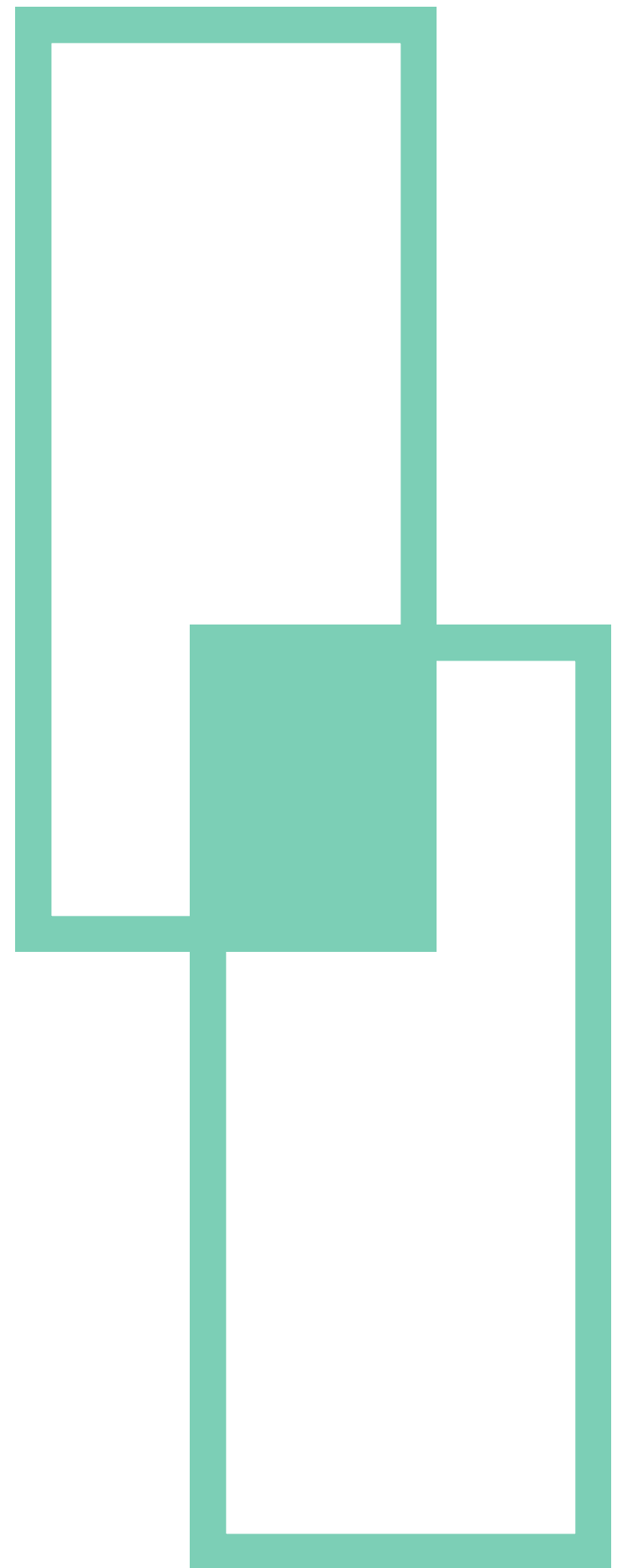
$$\stackrel{\pm}{=} \min_{\mathbf{a}} K(\mathbf{o} \mid \mathbf{a}) \stackrel{\pm}{=} K(\hat{\mu}^*)$$



Coding theorem

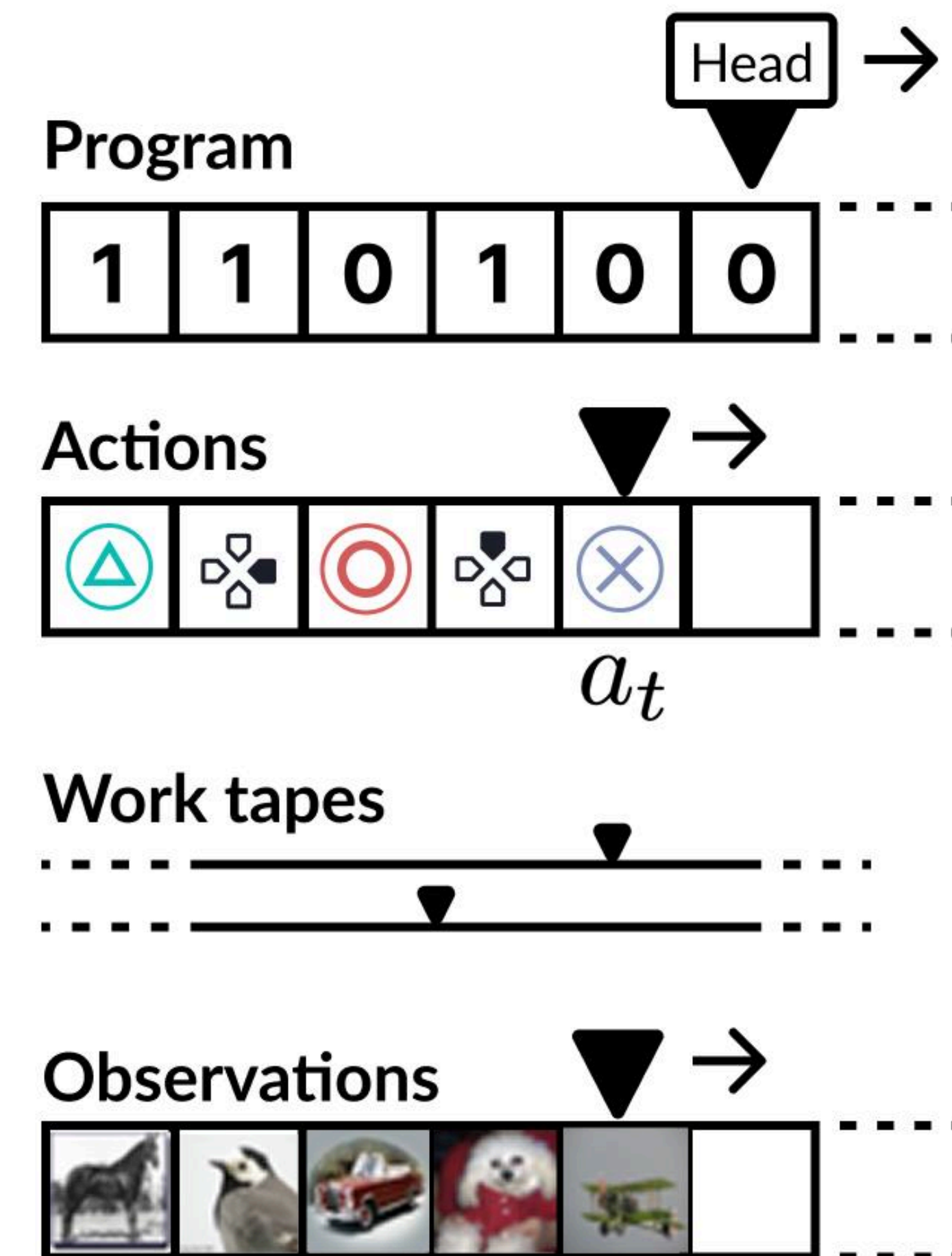


Niches are minimal programs lemma



Escaping The Dark Room Problem

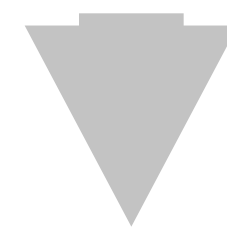
- Can we safely assume that the true environment has a lower bound on the complexity of the observation sequence? Recall that the environment is modeled after a Turing Machine.
- We cannot know a priori whether there is no action sequence such that $K(\mathbf{o} \mid \mathbf{a}) = O(1)$ (with $\mu(\mathbf{a}) = \mathbf{o}$) is a small integer.
- Is there even an environment μ such that:
 $\forall \mathbf{a} : K(\mathbf{o} \mid \mathbf{a}) \geq O(1)$ with $\mu(\mathbf{a}) = \mathbf{o}$
- If so, it would be interesting to detail how this environment would look and which additional assumptions do we need.



Incompressible Environments

An "incompressible" environment μ described by the program i^+ej is a program which runs and adds together (or combines) the outputs of an action dependent program $j(a) = o_j$ and an action independent minimal program $e(a) = o_e$

$$U(i^+ej, a) = \underbrace{o_e}_{\text{Incompressible Information}} + o_j = o = \mu(a)$$



$$\forall a \quad K(e) - K(a) - \ell(i^-j) \leq K(o \mid a)$$

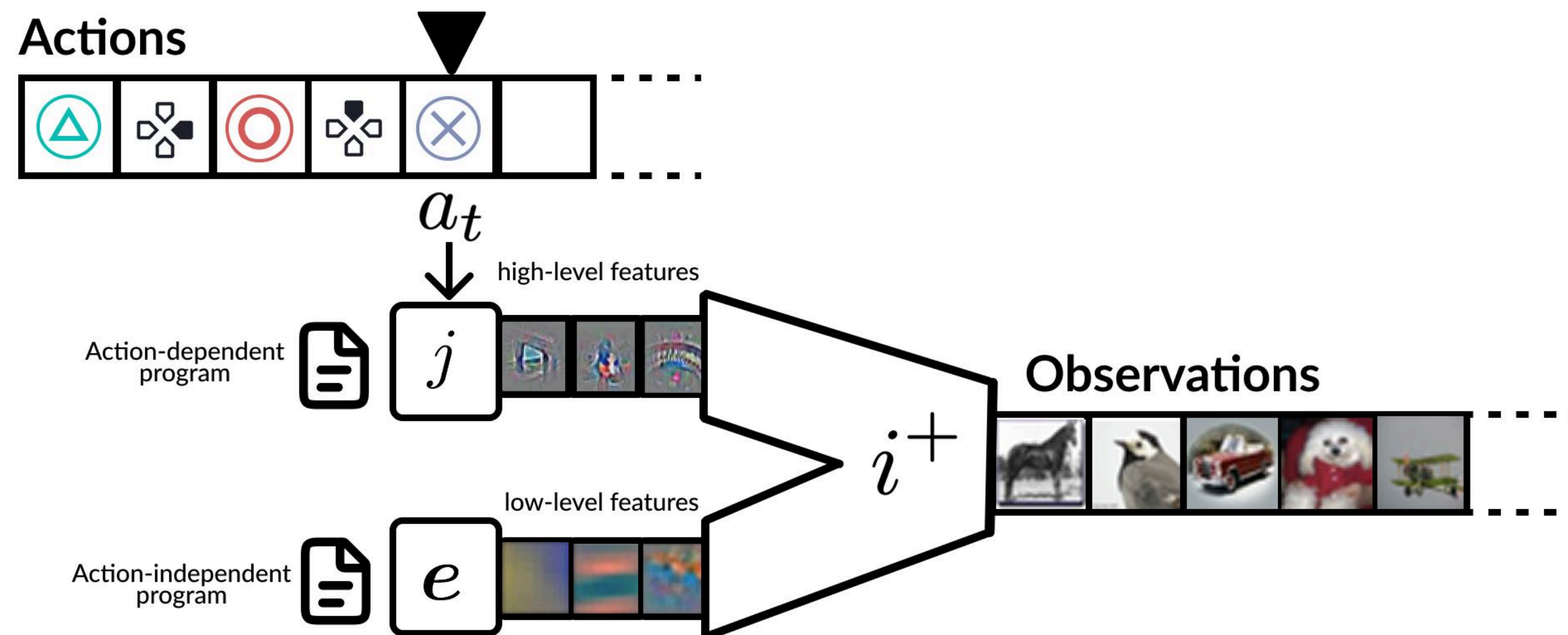
Naturalistic Assumption

$$m|\mathcal{O}| \geq m|\mathcal{A}|$$

Incompressible Environments

An "incompressible" environment μ described by the program i^+ej is a program which runs and adds together (or combines) the outputs of an action dependent program $j(a) = o_j$ and an action independent minimal program $e(a) = o_e$

$$U(i^+ej, a) = o_e + o_j = o = \mu(a)$$

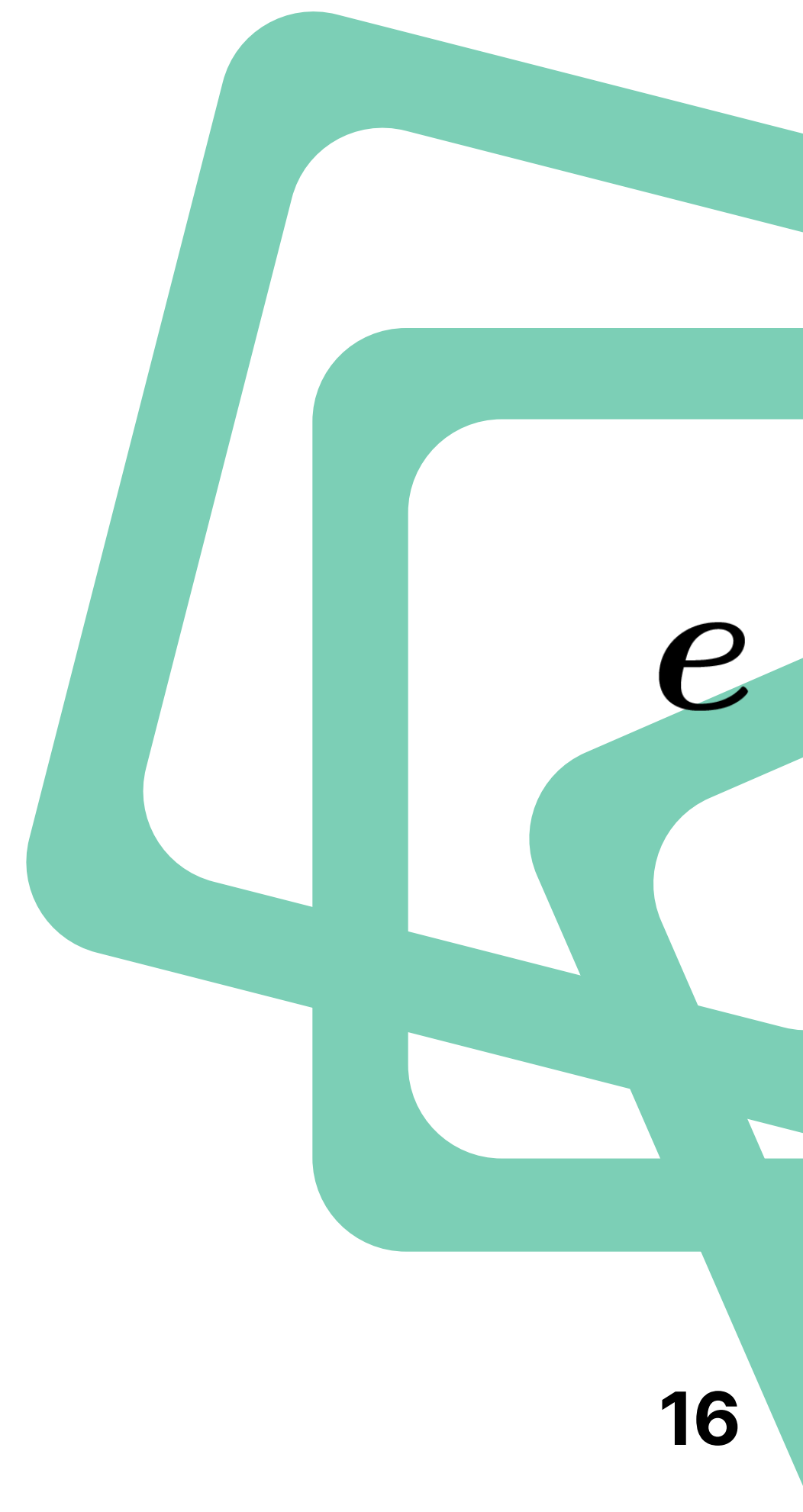


Incompressible Environments

Proposition: Every environment can be described/decomposed with an increase in complexity bounded by $K(u) + O(\log|\mathcal{O}|)$

$$\forall \mu \in \mathcal{M}, \exists u := i^+ e j :$$

$$\forall a : \mu(a) = u(a) \quad \text{and} \quad K(\mu) + O(\log m) + O(\log|\mathcal{O}|) \stackrel{+}{\geq} K(u)$$



The Agnostic Agent and its Regret

The oracle agent serves as a lower bound on surprise and exploration of the agnostic agent:

$$S_{\mu||\xi}^{\pi_{\mu||\xi}} \leq S_{\mu||\xi}^{\pi_{\xi||\xi}} \leq K(\mu)$$

The relationships above entail sublinear regret:

$$\begin{aligned} \text{Regret}_m(\pi_{\xi||\xi}, \mu) &= S_{\mu||\xi}^{\pi_{\xi||\xi}, m} - S_{\mu||\xi}^{\pi_{\mu||\xi}, m} \\ &\leq K(\mu) = o(m). \end{aligned}$$

Asymptotic Optimality

The surprise-minimizing agent is minimizing a quantity related to the per-step semi-measure loss.

$$\begin{aligned}
 \min_{a_t} \underbrace{1 - \sum_{o_t} \xi(o_t \mid h_{<t}a_t)}_{\text{semi-measure loss}} &= \min_{a_t} -\log \sum_{o_t} \xi(o_t \mid h_{<t}a_t) \\
 &\leq \min_{a_t} \sum_{o_t} \frac{\xi(o_t \mid h_{<t}a_t)}{Z_{a_t}} [-\log \xi(o_t \mid h_{<t}a_t)] \\
 &= \min_{a_t} \frac{S_{\xi \parallel \xi}^{\pi, 1}}{Z_{a_t}}
 \end{aligned}$$

Discussion



Summary: We have detailed the assumptions on which the agentic model of FEP might have to rely, and characterized the interesting properties of surprise minimization within UAI.

- **Reward specification:** Additional in-domain knowledge or other preferences may be incorporated in the prior (e.g. with a different reference UTM)
- **Safety considerations:**
 - Surprises minimization is tied to control of the surroundings
 - Nonetheless, a surprise-minimizing ASI might tend to settle for local optima instead of achieving its goals “no matter what”.
 - Indeed, it is an objective which favors maintaining the state of affairs.
- **Computable approximation:** it would be interesting to explore surprise-minimization in computable agents, since FEP models precisely bounded agents.

